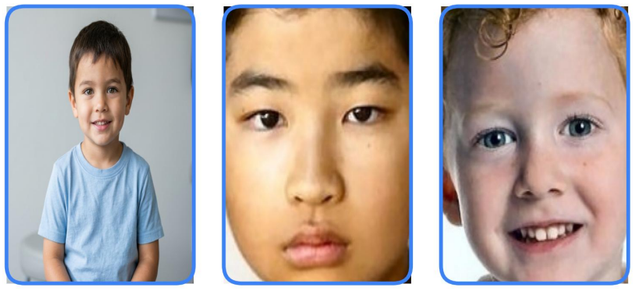
CNN-Based Autism Spectrum Disorder (ASD) Detection Pipeline

Reverse-C Sequential Flow: Data Ingestion → Facial Preprocessing → Dataset Splitting → CNN Architecture Comparison → Clinical Evaluation

01 Data Collection

Diverse Multi-Subject Pediatric Cohort



Subject 01 | Subject 02 | Subject 03

Pediatric Dataset: [Pending Intake] Images

Raw Photos →

02 Face Preprocessing

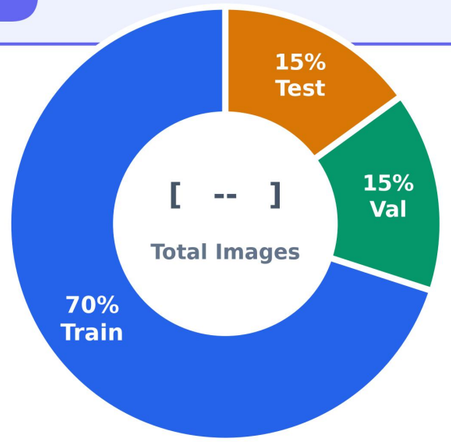
Automated Landmark Detection & Cropping



MTCNN Detect & Align (224 × 224)

Aligned 224 × 224 →

03 Dataset Splitting



Train (70%)	Val (15%)	Test (15%)
[--] imgs	[--] imgs	[--] imgs

↓ Stratified 70/15/15 Splits

04 Model Training & Comparison

6 CNN CANDIDATE ARCHITECTURES (CROSS-ENTROPY LOSS, ADAM)

ResNet-50	Pending	[-- %]	Deep Residual (50-Layer)
ResNet-34	Pending	[-- %]	Residual Baseline (34-Layer)
VGG-19	Pending	[-- %]	Deep Feedforward (19-Layer)
VGG-16	Pending	[-- %]	Standard Baseline (16-Layer)
MobileNet-V2	Pending	[-- %]	Lightweight Inverted Residuals
AlexNet	Pending	[-- %]	Classic Convolutional Network

Optimal Backbone: [To Be Selected Based on Validation Results]

← Selected Backbone

05 Evaluation & Diagnosis

CLINICAL SCREENING OUTCOME

Binary Classification: ASD vs. Neurotypical Control

Predicted Label: [Pending] | Model Confidence: [-- %]

Overall Test Accuracy	To Be Measured	[-- %]
AUROC Discriminability	To Be Measured	[-- %]
Diagnostic F1-Score	To Be Measured	[-- %]
Sensitivity / Specificity	To Be Measured	[-- / --]

Clinical Decision Support Ready • Target Latency: [--] ms